



Design project 2

RESO: ENHANCING SOCIAL COMMUNICATION FOR THE DEAF & HARD-OF-HEARING

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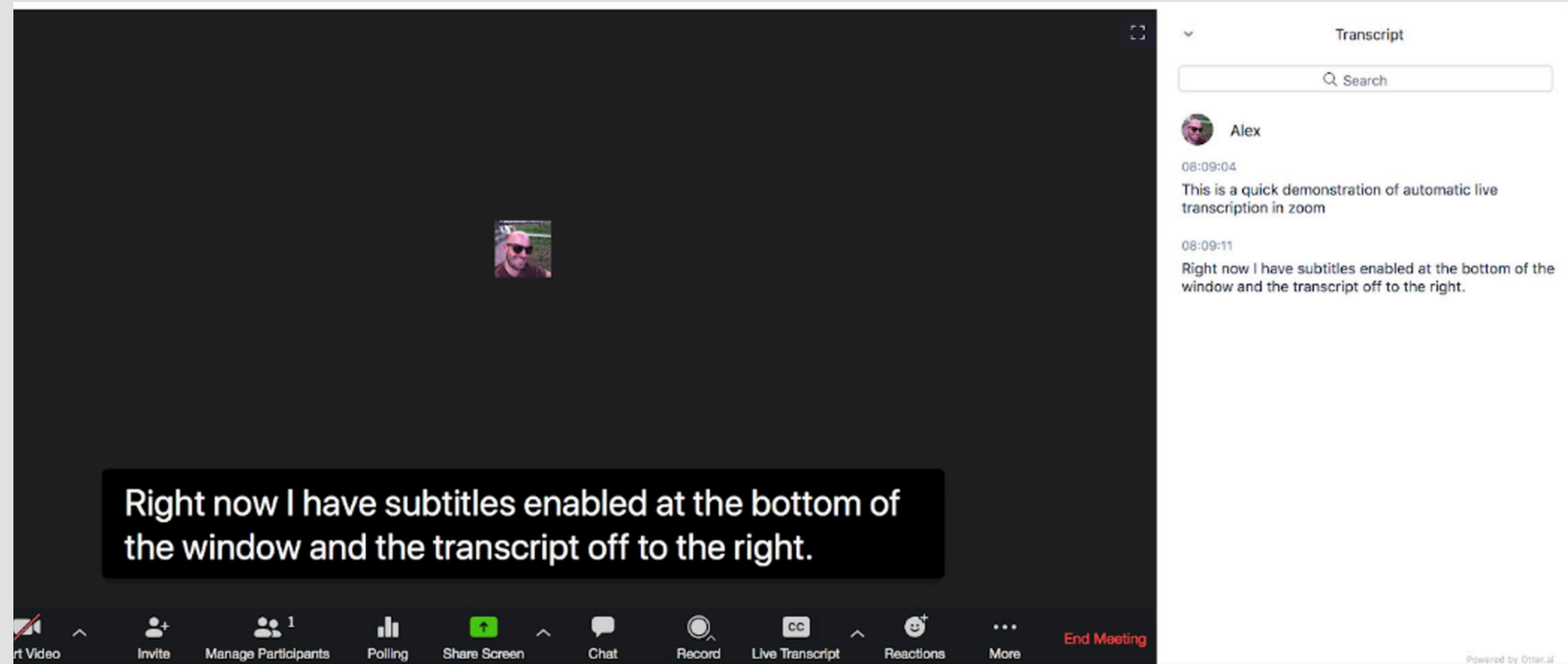
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Background

Currently more than 1.5 billion people live with hearing loss;
430 million of them have disabling hearing loss.

Online meetings
improved accessibility
through captions, but
still fail to convey
emotional and social
context for DHH users.



Introduction

Text-only captions flatten communication and increase cognitive load during online interactions.



RQ

What is the effect of combining emotion-coded text and visual sounds rhythms on cognitive load and emotional understanding of DHH users in online meetings in comparison to traditional text captions?

Introduction

Visualize the emotion and the rhythm of the speaker's voice

Baseline

"I spoke to her last week, actually. She seemed fine."



New

"Honestly I don't even know what to say right now."

VOICE INTENSITY



The DHH Meeting Story

The Unseen Barrier

The Hearing Barrier:

It is not a lack of words, but a lack of nuance. Hearing coworkers pick up on a "tone of voice" that instantly signals sarcasm, frustration, or support. DHH users receive only the literal text, leaving them "emotionally blind" to the true office climate.

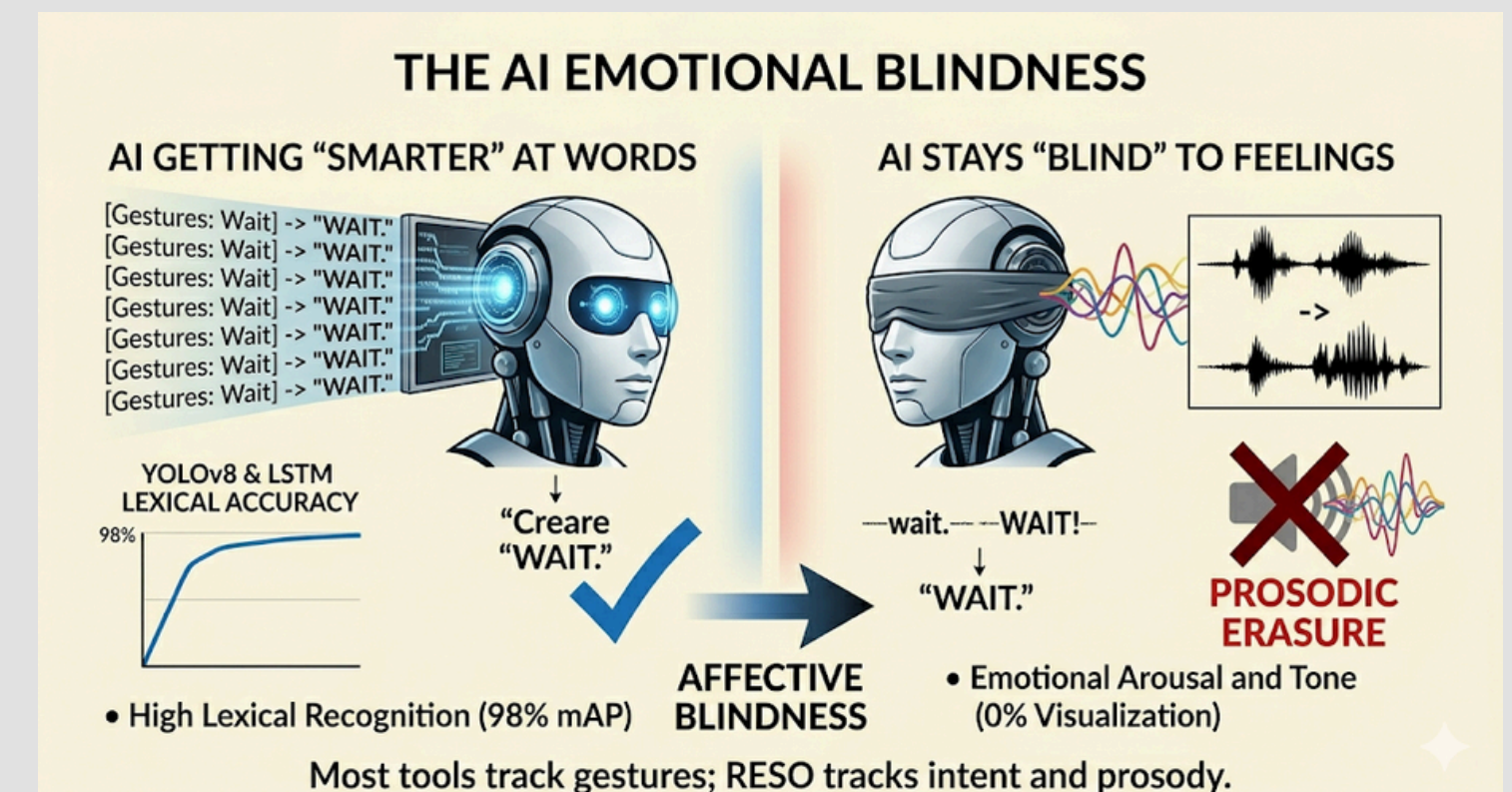
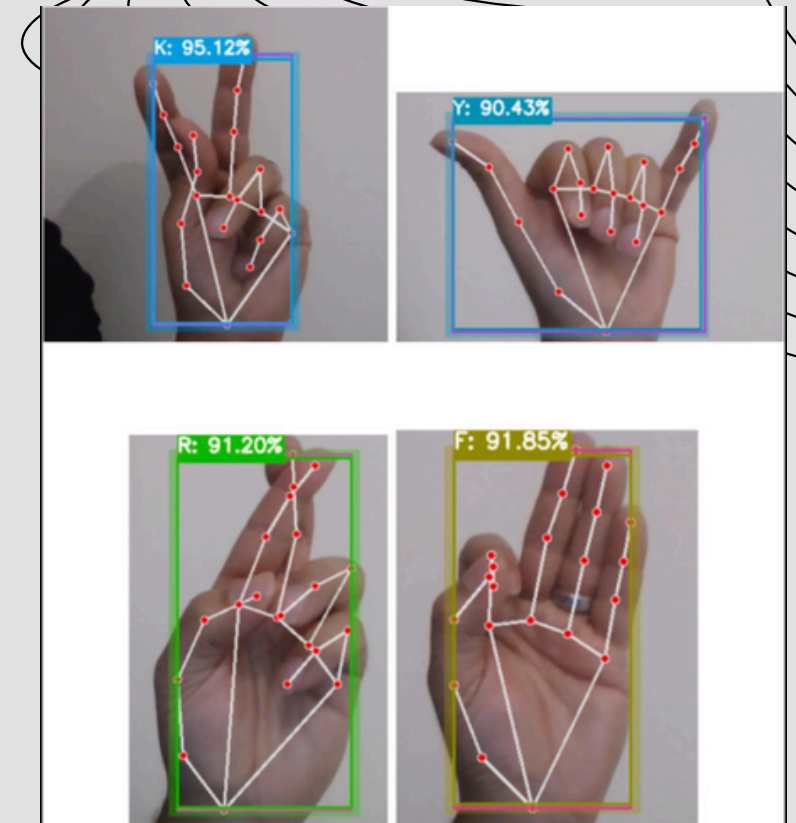
The Office Environment:

In fast-paced professional meetings, turn-taking is chaotic. The "Flat" experience of standard captions makes it impossible to sense the urgency or tempo of a debate, leading to "Social Lag" where the DHH professional is always one step behind the social flow.

The Tech Lands

High Accuracy, Low Empathy

- **The Translation Bias:** Current AI models (CNNs, LSTMs, YOLOv8) focus 99% on lexical accuracy getting the right word—while ignoring prosody (tone, pitch, and energy).
- **Gestures -> Intent:** We can map a hand shape to the word "Wait," but we can't map the vocal urgency of someone screaming "WAIT!" vs. a calm "wait."
- **The "Text-Only" Trap:** Technological progress has created a standard where "accessibility" just means "more text," which doesn't solve the social gap.



Source: Pataca et al. (2023); Kim et al. (2023); ZainEldin et al. (2024)

Meeting Breakdown

Limitations of Current Tools

Cognitive Overload

Linear captions can't handle multiple speakers, confusing the user.

The "Word Cloud" Failure

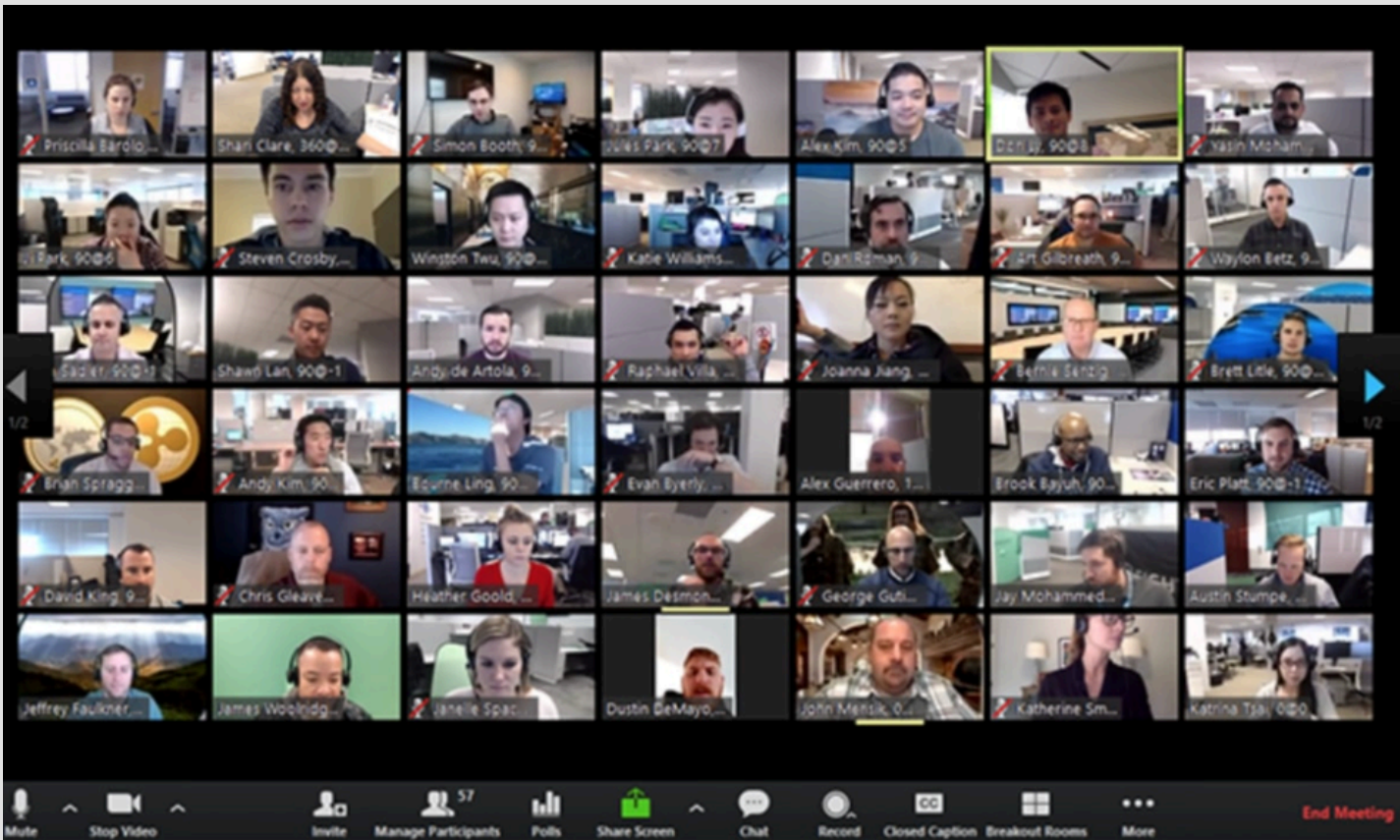
Static topics lose the "live" urgency and flow of conversation.

HMD Limitations

Headsets fail to pinpoint exactly who is speaking on screen.

Visual Dispersion

Eye-jumping between faces and text causes extreme fatigue.



Call-out: image_comp.jpg
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[image_comp.jpg]

**MEETING BREAKDOWN:
THE STANDARD CAPTION FAILURE**

Call-out: image_comp.jpg
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e= evaluation.jpg]

**BARRIER 1:
MULTIPLE SPEAKERS**

**BARRIER 2:
VISUAL DISPERSION**

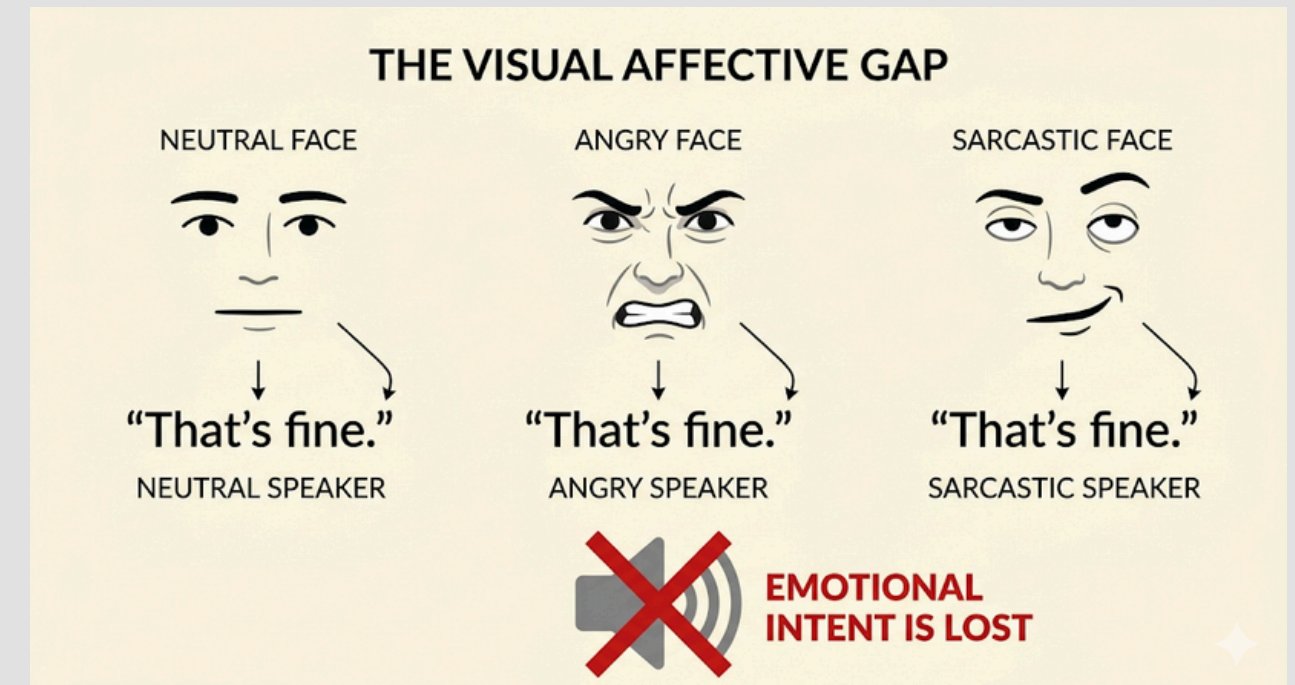
[SINGLE CAPTION STREAM] "Wait... I can't... overlap... sorry... no, you go..."

[SINGLE STREAM CAPTION] "Yes, let me... you go... okay, so, wait, I can't..." [CONFUSION]

Research Gap

The "Affective Blindness" of Visual Outputs

- **Lexical vs. Prosodic:** Current AI prioritizes lexical accuracy (the words) over prosodic visualization (tone, intensity, and rhythm).
- **Emotional Erasure:** Standard monochromatic captions flatten speech, erasing the subtle distinctions between excitement, anger, or sarcasm.
- **Visual Homogeneity:** Whether a speaker whispers or screams, the resulting visual text looks exactly the same, creating a massive interpretive burden for DHH users.
- **The Intent Gap:** Standard tools do not externalize a speaker's emotional arousal into a distinct visual aesthetic.



Design process

Low-fidelity prototype

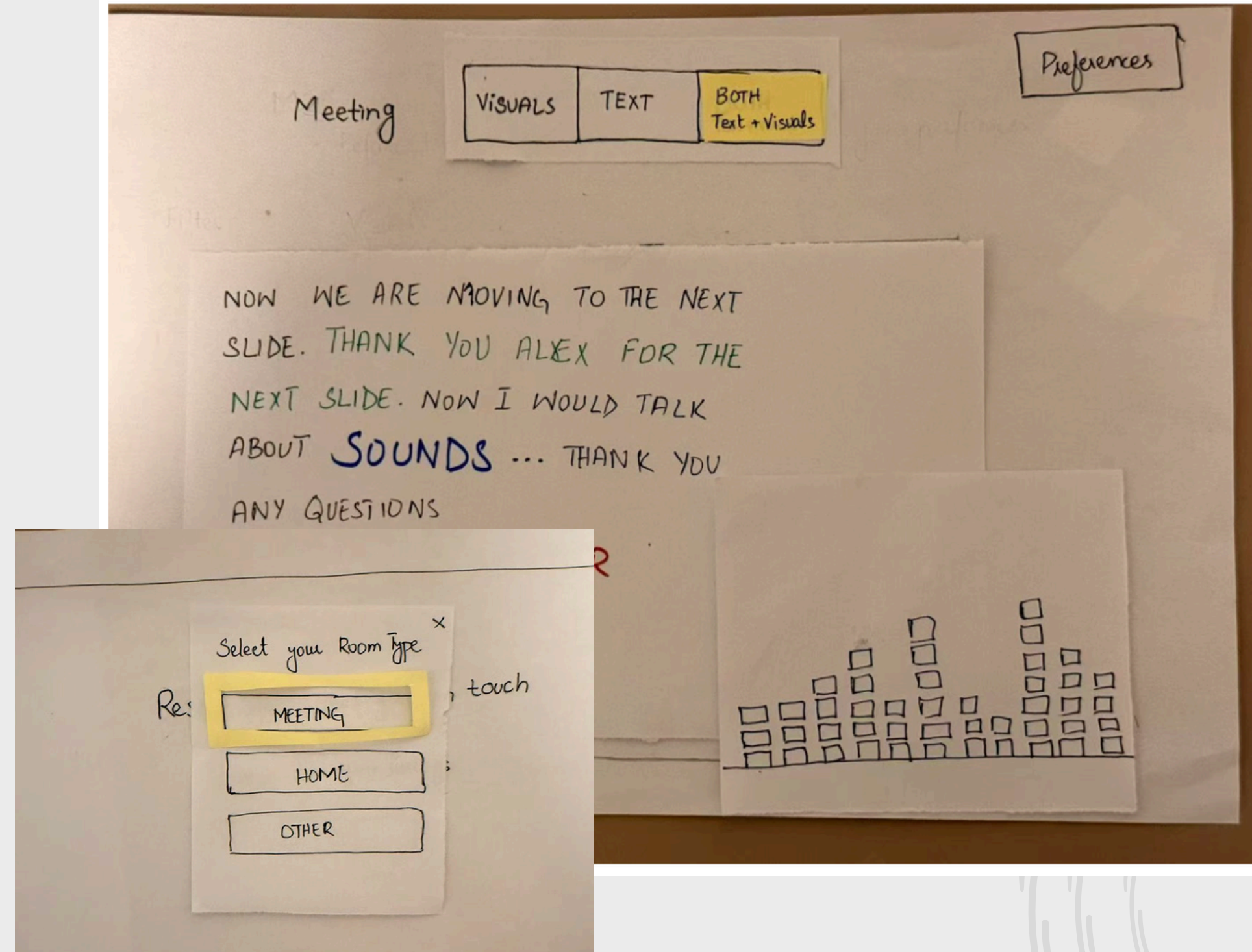
Concept:

A screen-based tool for Online Meetings.

Key Features:

- Visual Rhythm: Used "Square Blocks" to show sound frequency.
- Emotion Colors: Text changes color (e.g. Red = Angry) to show feelings.

The screen display includes not only text but also visual effects.



Design process

Visual Rhythm(Square Blocks)

Concept:

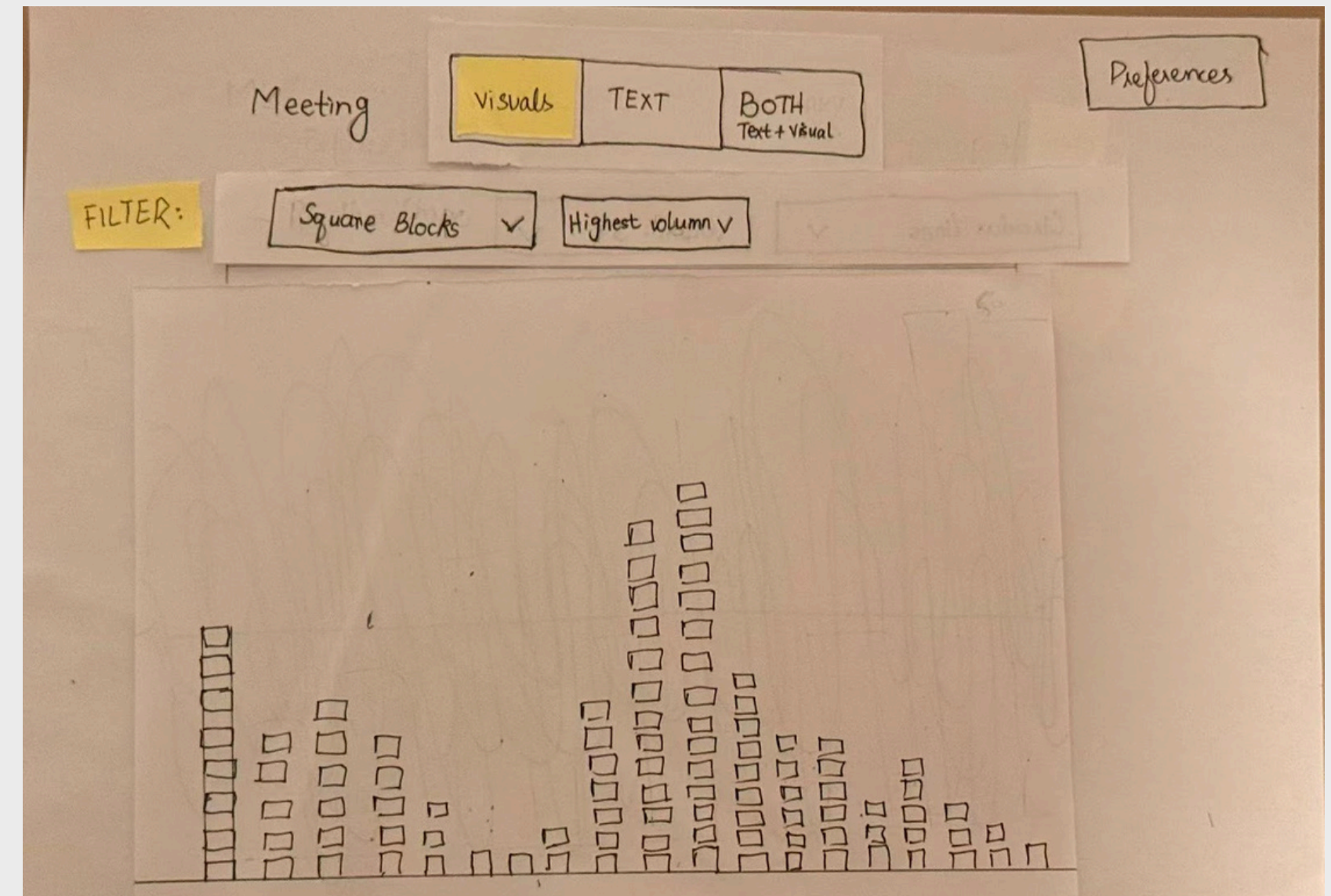
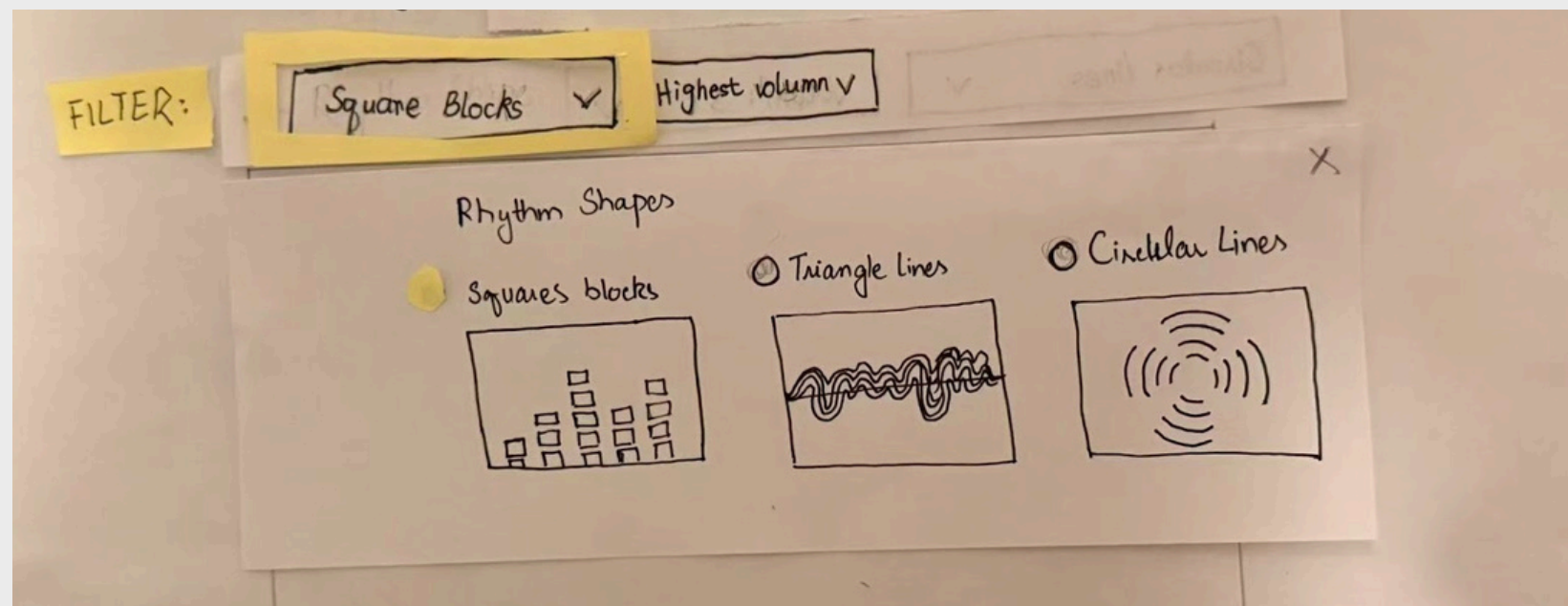
Turning abstract sound into visible shapes.

Feature:

Use "Square Blocks" to show sound frequency

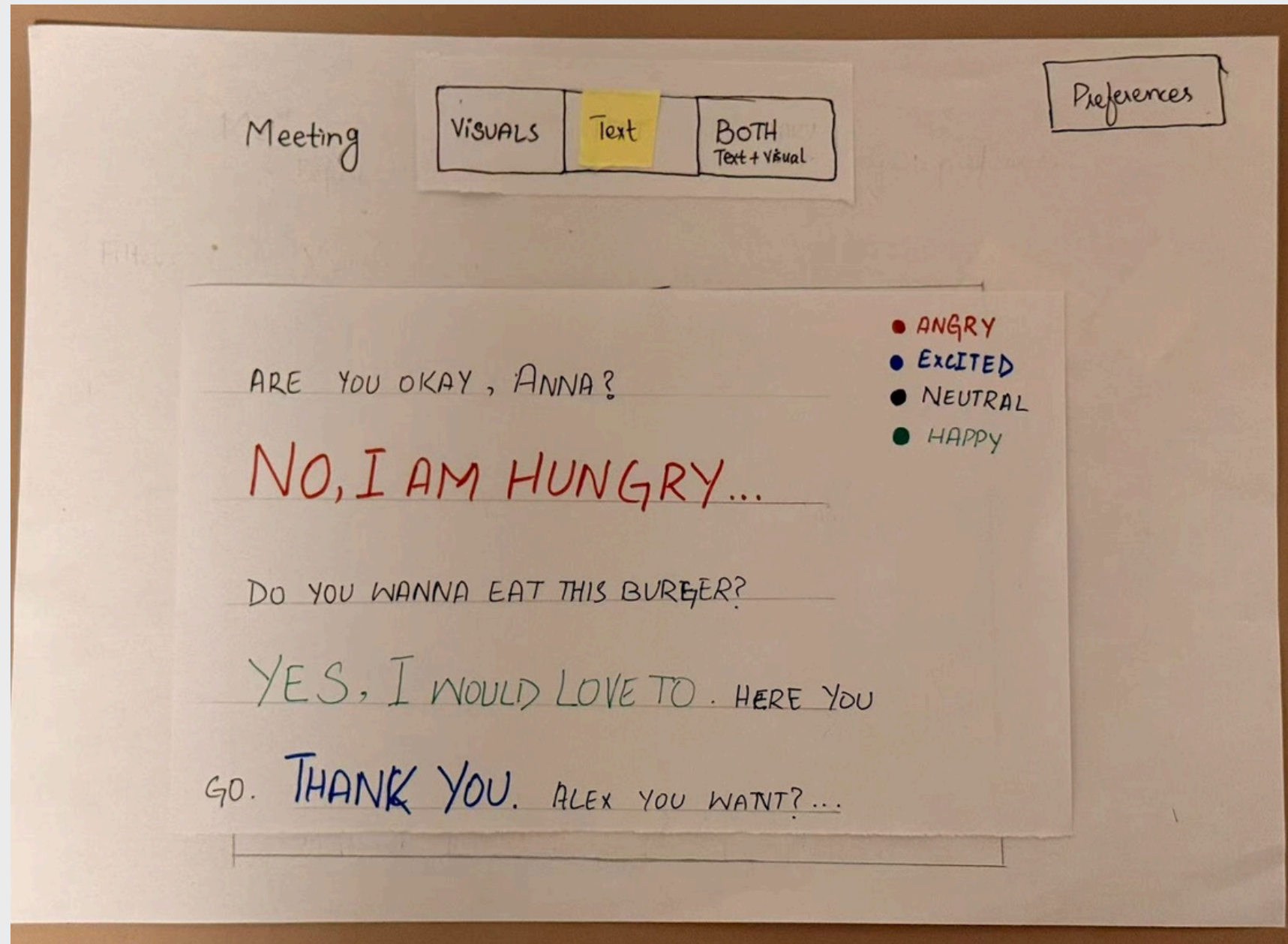
Benefit:

Users can feel the "energy" and "intensity" of the conversation.



Design process

Emotion Colors (Feelings in Text)



Problem:

Traditional captions lack emotional tone.

Solution:

Emotion Colors (Text changes color to show feelings).

Benefit:

- Understand emotion instantly.
- Reduce cognitive load (no more guessing tone).

High-fidelity System

Browser Speech Capture

This page captures microphone speech in the browser and forwards final transcripts into the local overlay app running on this machine.

Speech recognition is available in this browser.

Listening. Final transcript sent to overlay.

Listening: I'm very

LIVE SOUND RHYTHM

Posting captions to <http://127.0.0.1:4153/caption>

LIVE 35% · INTERRUPT 0%



I'm very angry

ANGER

I'm sad

SADNESS

- Keep this page open while the Electron overlay app is running.
- Allow microphone access when the browser asks.
- Final recognized phrases will be sent to the overlay automatically.

High-fidelity System

Emotion Speech Overlay

Server live on 4153

Drag anywhere in the top bar to move this window.

Speech Input

Microphone idle

Click Start Listening and allow microphone access.

No live transcript yet.

Start Listening

Stop Listening

Open Browser Capture

Clear Conversation

Speech recognition is available in this runtime, but Electron can still be unreliable. Browser Capture is the safer fallback.

Display Settings

Theme

LIVE 10% · INTERRUPT 0%

LIVE SOUND RHYTHM

Posting captions to <http://127.0.0.1:4153/caption>

LIVE 35% · INTERRUPT 0%

Start Listening

Stop Listening

I'm very angry

ANGER

I'm sad

SADNESS

• Keep this page open while the Electron overlay app is running.

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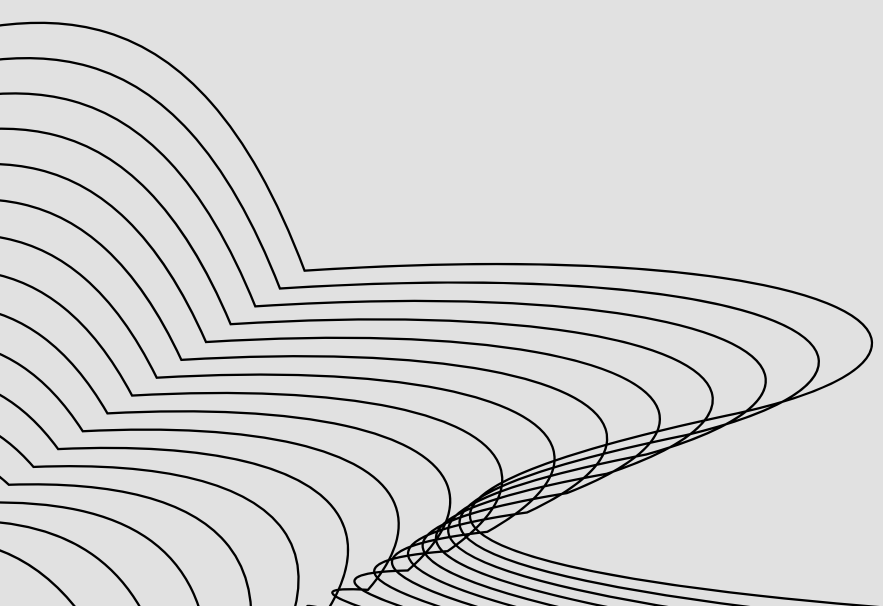
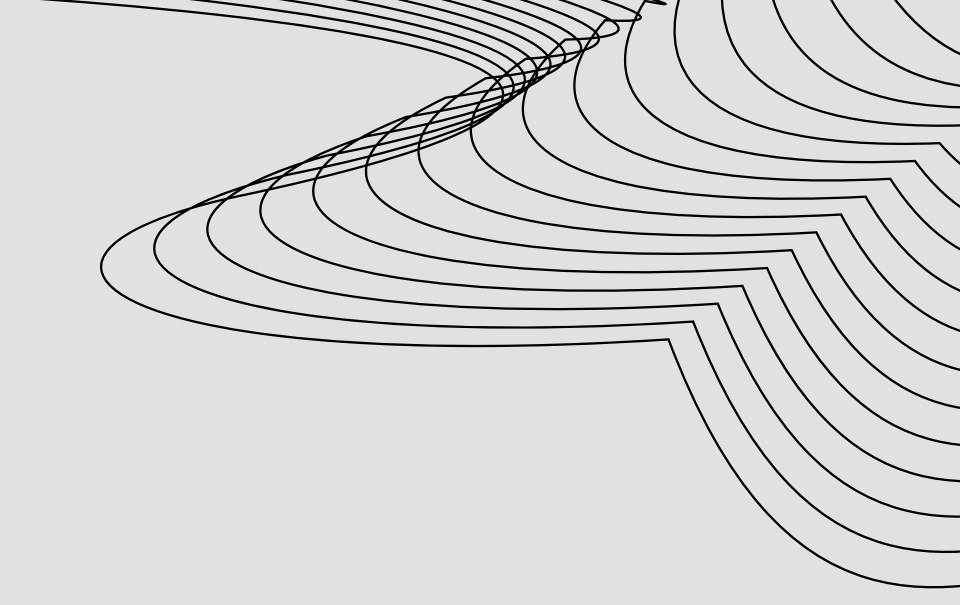
I'm very angry

I'm sad

ANGER

SADNESS

S



DEMO

Emotion Speech Overlay

Server live on 4153

Drag anywhere in the top bar to move this window.

Speech Input

Microphone idle

Click Start Listening and allow microphone access.

No live transcript yet.

Start Listening

Stop Listening

Open Browser Capture

Clear Conversation

Speech recognition is available in this runtime, but Electron can still be unreliable. Browser Capture is the safer fallback.

Display Settings

Theme

LIVE SOUND RHYTHM

WinterSchool202
6_DHH.pdf

LISTENING STANDBY

3m.bgcode

HYPOTHESES

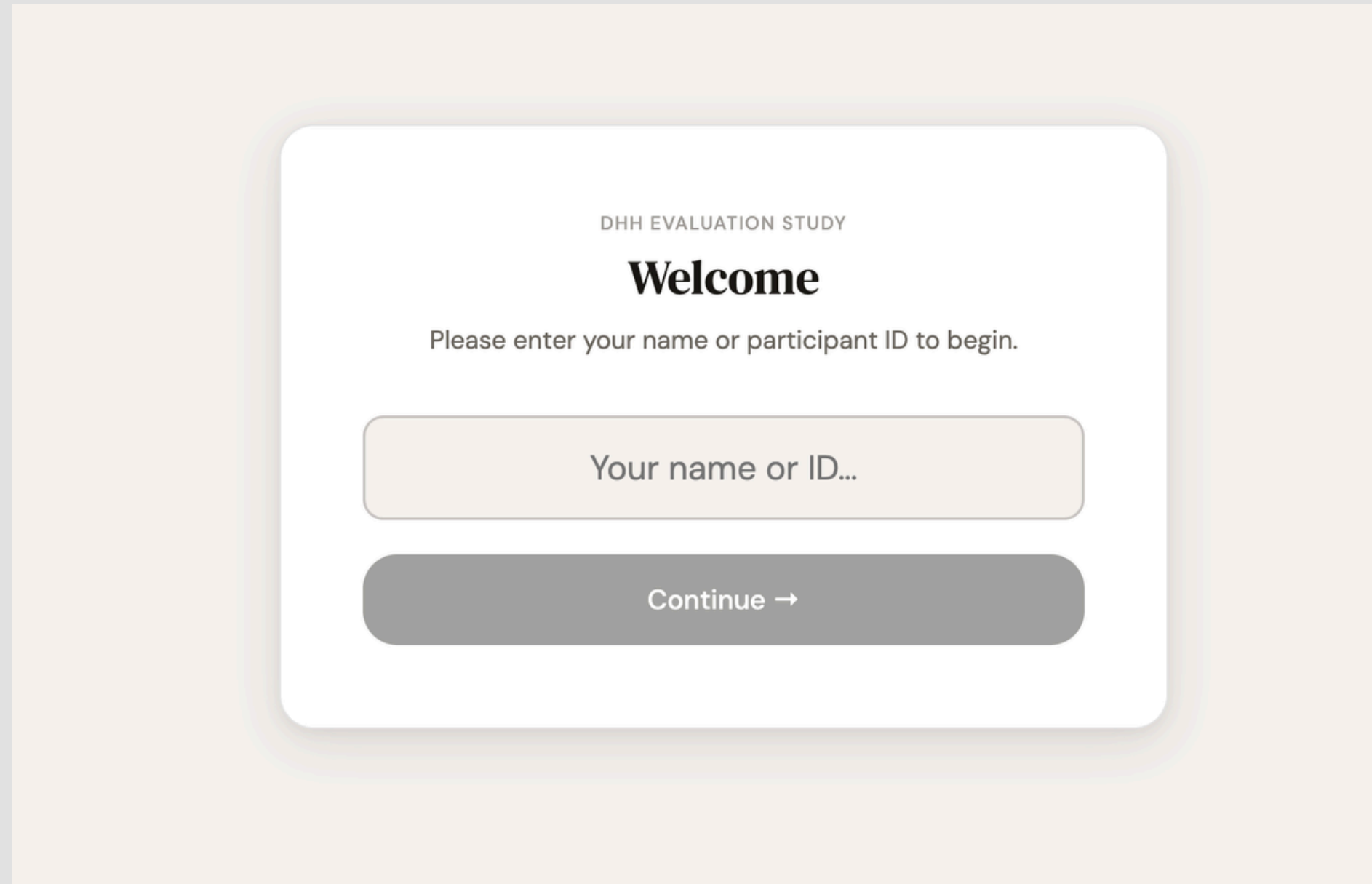
H1

The emotion encoded text will greatly enhance the accuracy or speed of the user in recognizing the tone of the speaker in terms of emotion as opposed to the normal flat captions.

H2

By employing visual rhythms to display the volume of sound and interruption, the overall cognitive load of the users will be decreased in comparison with the conditions when only the text is read.

Evaluation Phase



DHH EVALUATION STUDY

Welcome

Please enter your name or participant ID to begin.

Continue →

<https://danietzio.github.io/deafDHH.github.io/>

Evaluation Phase

Participant Onboarding: Participants must provide informed consent by acknowledging their voluntary participation and agreeing to the use of anonymized data before evaluating Reso's emotion-coded captions.

Ethical Framework: The study ensures participant safety by disclosing minimal visual fatigue risks, guaranteeing pseudo-anonymized data security, and maintaining a policy of voluntary withdrawal at any time.

BEFORE WE BEGIN

Consent Form

Please read carefully and tick all boxes to proceed.

Study Title

Evaluating the impact of visual sound rhythms and emotion-coded captions in online meetings for Deaf and Hard-of-Hearing users.

Purpose

This study examines a digital accessibility tool designed to help Deaf and Hard-of-Hearing (DHH) users better understand the emotional tone of speech during online meetings. We compare standard text captions against a new interface that uses color-coded text and visual

- ☐ I have read and understood the information above, and I agree to participate.
- ☐ I understand my participation is voluntary and I can withdraw at any time.
- ☐ I consent to my anonymised data being used for research purposes.

Decline

I Agree & Continue →

Evaluation Phase

Task Walkthrough: Participants are provided a comprehensive overview of the study flow, which includes watching short video clips and identifying the perceived emotion of the speaker as quickly as possible.

THE TWO INTERFACES

Condition
A

Standard captions

Plain white text captions appear below the video. No colors or additional visual cues.

Condition
B

Emotion-coded captions + sound rhythm

After you respond, caption text is revealed in a color matching the detected emotion. A live frequency graph shows voice intensity throughout.

BEFORE WE BEGIN

How this study works

Please read through everything before starting.

WHAT YOU WILL DO

1

Watch video clips

You will watch a series of short video clips — one sentence at a time. Each clip shows a person speaking a single sentence.

2

Identify the emotion

After each sentence finishes, select the emotion you perceived as quickly and honestly as possible. Your reaction time is measured from when the video ends until you make your selection.

3

Two rounds of 6 sentences

You will complete two blocks with a short break in between. Each block uses a different interface.

4

Short questionnaire after each block

After each block you will fill in a brief workload questionnaire (NASA-TLX). This takes about 2 minutes.

Evaluation Phase

Simulated Accessibility Experience: To simulate the experience of Deaf or Hard-of-Hearing users, participants are shown muted video clips where they must rely entirely on visual and textual cues.

Emotion Recognition Task: Users are required to select the speaker's perceived emotion from six distinct categories: Happy, Angry, Sad, Surprised, Neutral, and Fearful.

Performance Tracking: The interface captures the user's reaction time and accuracy to measure how effectively the captions communicate emotional intent.

Sentence 1 of 6



"Oh, wow. That's... that's really something."

Reaction time 3.69 s

What emotion did you perceive from the speaker?

 Happy	 Angry	 Sad
 Surprised	 Neutral	 Fearful

Confirm & Next →

Evaluation Phase

Subjective Workload Rating: Participants use the NASA Task Load Index (NASA-TLX) to rate their perceived cognitive demand across dimensions such as mental, physical, and temporal demand after each test block

Raw TLX score (unweighted average)

50

Lower = less cognitive load

Save & Continue →

CONDITION A — WORKLOAD RATING

Workload Assessment

NASA Task Load Index — rate each dimension for the block you just completed.

Mental demand

How much thinking, deciding, or calculating was required?

Very lowVery high

50

Physical demand

How much physical activity was required?

Very lowVery high

50

Temporal demand

How much time pressure did you feel?

Very lowVery high

50

Performance

How successful were you in accomplishing the task?

PerfectFailure

50

Effort

How hard did you have to work to accomplish your performance?

Very lowVery high

50

Evaluation Phase



Short break

The second interface will begin shortly.
Take a moment to rest your eyes.

0:04

The next block starts automatically, or click below when
ready.

Begin Condition 2 →

Evaluation Phase

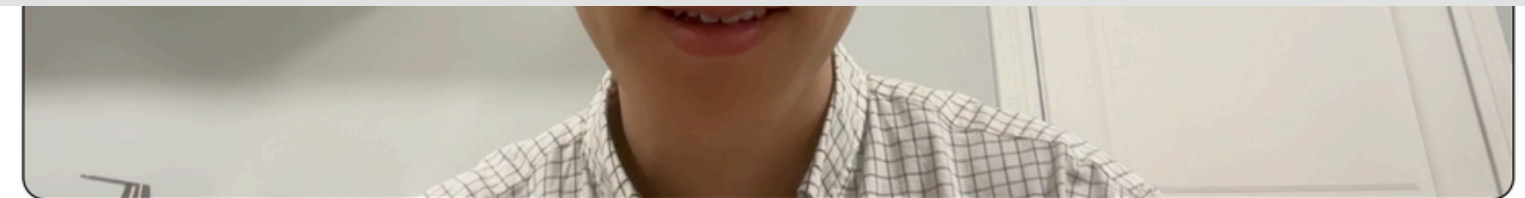
Condition B

Sentence 1 of 6



"Honestly I don't even know what to say right now."

VOICE INTENSITY



"Honestly I don't even know what to say right now."

VOICE INTENSITY



Reaction time

4.11 s

What emotion did you perceive from the speaker?



Happy



Angry



Sad



Surprised



Neutral



Fearful

Confirm & Next →

Evaluation Phase

Subjective Workload Rating: Participants use the NASA Task Load Index (NASA-TLX) to rate their perceived cognitive demand across dimensions such as mental, physical, and temporal demand after each test block

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How successful were you in accomplishing the task?

PerfectFailure

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Effort

How hard did you have to work to accomplish your performance?

Very lowVery high

50

Evaluation Phase

Debrief Questions

Qualitative Feedback Collection:

Following the performance tasks, participants complete a debrief session to provide subjective insights on interface preference and the perceived helpfulness of the visual cues.

FINAL STEP

Debrief questions

Please answer these 3 short questions. Your responses are very valuable.

1. Which interface felt easier to follow, and why?

Your answer...

2. Did the emotion colors or the frequency graph help you understand the speaker? Were there any moments where they felt misleading?

Your answer...

3. Is there anything you would change or add to the experimental interface?

Your answer...

Submit & Finish →

Results Analysis

ACCURACY — CONDITION A

50.9%

Standard captions

ACCURACY — CONDITION B

63.0%

Reso interface

ACCURACY DIFFERENCE (B – A)

+12.0%

p < .05 ✓ Significant

MEAN RT — CONDITION A

3.73s

Standard captions

MEAN RT — CONDITION B

4.21s

Reso interface

RT DIFFERENCE (B – A)

+0.47s

p = 0.589 ✗ Not significant

NASA-TLX TOTAL — A

39.1

Standard captions

NASA-TLX TOTAL — B

44.1

Reso interface

NASA-TLX DIFFERENCE (B – A)

+5.1

p = 0.264 ✗ Not significant

Results Analysis



Figure 3. Emotion recognition accuracy (%) broken down by emotion type for Condition A and B. Differences reveal which emotional cues benefited most from Reso's visual enhancements.

Emotion	Accuracy A Standard	Accuracy B Reso	Difference (B – A)
Happy	66.7%	72.2%	+5.6%
Angry	27.8%	61.1%	+33.3%
Sad	33.3%	61.1%	+27.8%
Surprised	77.8%	72.2%	-5.6%
Neutral	77.8%	83.3%	+5.6%
Fearful	22.2%	27.8%	+5.6%

Results Analysis

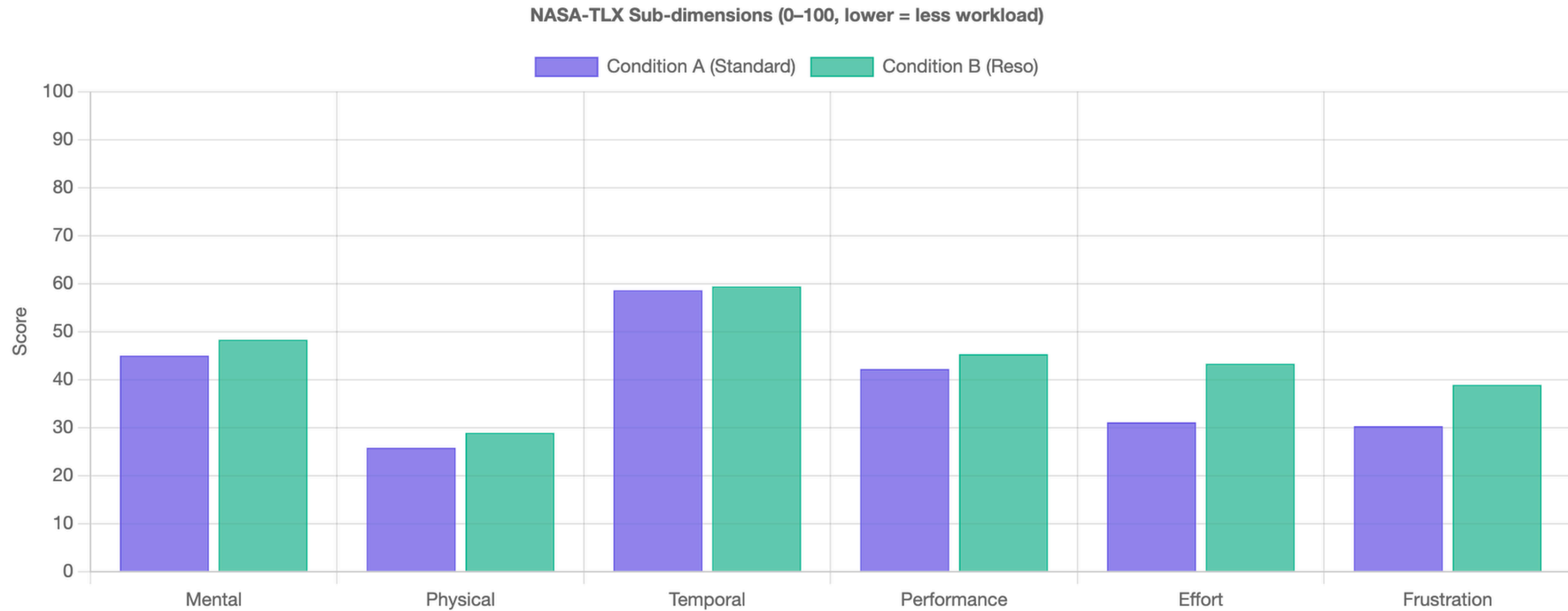


Figure 4. Mean NASA-TLX scores across all six dimensions for Condition A and B (scale 0–100, lower = less workload). Overall cognitive load was higher under Condition B (Reso).

Results Analysis

ACCURACY — CONDITION A

50.9%

Standard captions

ACCURACY — CONDITION B

63.0%

Reso interface

ACCURACY DIFFERENCE (B – A)

+12.0%

p < .05 ✓ Significant

MEAN RT — CONDITION A

3.73s

Standard captions

MEAN RT — CONDITION B

4.21s

Reso interface

RT DIFFERENCE (B – A)

+0.47s

p = 0.589 ✗ Not significant

NASA-TLX TOTAL — A

39.1

Standard captions

NASA-TLX TOTAL — B

44.1

Reso interface

NASA-TLX DIFFERENCE (B – A)

+5.1

p = 0.264 ✗ Not significant

Qualitative Analysis

Q1 – Which interface felt easier to use?

The majority of participants found **Condition A (standard captions)** easier, citing **lower cognitive demand** and **fewer competing visual elements**.

Common reason for preferring A: participants could focus solely on facial expressions and text without processing extra visual layers.

Q2 – Did the emotion colors or frequency graph help identify the speaker's emotion?

Emotion colors were generally seen as helpful and intuitive — they reinforced caption meaning without adding confusion.

The frequency/sound rhythm graph was widely described as unhelpful or distracting. Most participants did not understand how to interpret it

Key issue: the graph lacked a legend or scale reference, making it uninterpretable for most users.

P4

If there are subtitles with color changes, it will be easier for me to understand what this person wants to express with the expression

P6

The frequency graph did help a bit, even though it was difficult to read and understand in a short time

P8

I'd say second one, the color addition was definitely a big plus, although the voice frequency was kinda useless and taking up a bit too much space

Qualitative Analysis

Q3 — What would you change or add to experimental design?

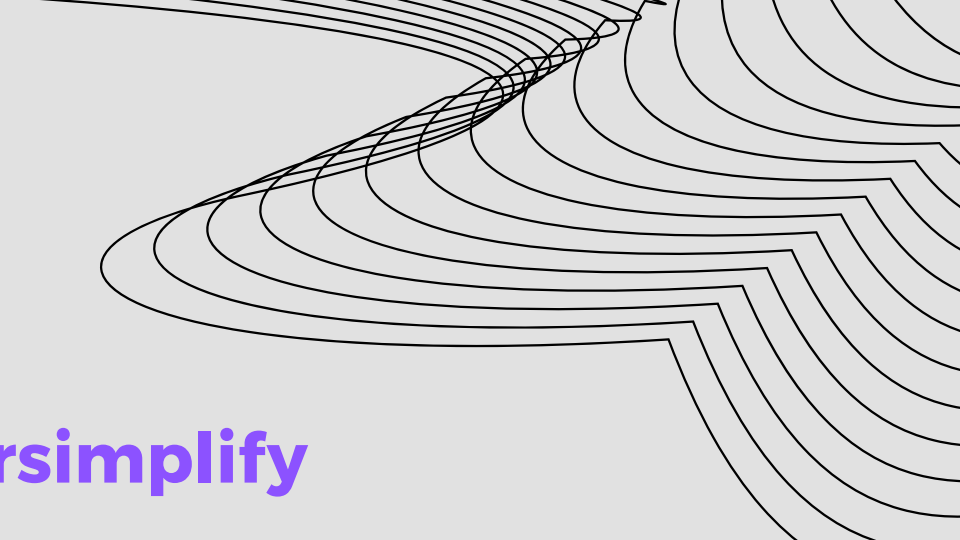
Frequency graph: Add a legend, axis labels, or a tutorial so participants understand what the waveform represents.

Limited Emotion options: Expand the set to include nuanced emotions like sarcasm, worry, or disappointment.

Color design: Make emotion colors more obvious and easily detectable (e.g., use as background rather than text color).

Timer: The visible countdown caused stress; consider hiding or softening it.

LIMITATIONS & DISCUSSION



Hearing-only sample

All 18 participants were hearing. Muting videos simulates, but cannot replicate, the DHH experience of social exclusion, caption fatigue, or learned coping strategies.

Isolated single sentences

Each video showed one speaker delivering one sentence. Real meetings involve overlapping speech, turn-taking, and multiple participants — far higher cognitive demands.

Muted video ≠ real meeting

Watching a pre-recorded muted clip removes social pressure, ambient noise, real-time interaction, and the need to manage attention across multiple streams.

Discrete categories oversimplify

Mapping affect to 4-6 colors flattens sarcasm, mixed emotion, and cultural color associations. The color schema was not validated with DHH users.

No Inputs from DHH. Community

Design decisions were made without participatory input from the DHH community, risking solutions shaped by hearing assumptions rather than Deaf cultural values.

FUTURE WORK



Study with actual DHH users

Recruit Deaf and Hard-of-Hearing participants in real online meetings (Zoom/Teams), measuring fatigue, social comfort, and perceived belonging over multiple sessions.

Live multi-speaker meetings

Test Reso in dynamic group conversations with overlapping speech, real turn-taking, and ambient noise to evaluate whether the visual cues scale beyond isolated sentences.

User-defined color mapping

Allow DHH users to assign their own emotion-color schema, respecting cultural differences, individual sensory preferences, and existing color blindness accessibility needs.

Size Change, Micro animations

For the next versions, we will explore possibility of using different sizes based on the sound frequency and also adding micro animations to make users more involved in the meeting.

AR & haptic prototype evaluation

Build on the AR headset concept (Prototype 2) with functional hardware and directional haptic motors, then conduct a formal comparative study against the laptop interface.

Image Recongition

We will explore having more accurate emotion detection by using image recognition added on top of the current application.

Conclusion

No More Flat Communication Experience

Traditional text-based captions frequently omit **critical emotional cues**, leading to a "**flat**" communication experience

Better Accessibility for DHH Community

This approach advances digital accessibility by shifting toward tools that prioritize Deaf culture and lived experience over simple speech-to-text translation

The image features a light gray background with the text "THANK YOU" centered in a bold, black, sans-serif font. In the four corners, there are decorative elements consisting of multiple concentric, wavy lines that create a sense of depth and movement, resembling stylized ripples or topographical contours. The lines in the top-left and bottom-right corners are thicker and more pronounced, while the lines in the top-right and bottom-left corners are thinner and more delicate.

THANK YOU